QUIZ ON NANOTECHNOLOGY

Q.1. What are nano materials?

a) material in which atom move away from each other and with size in the range of 1 to 50 nano meters.

b) material in which atom do not move away from each other and with size in the range of 1 to 100 nano meters.

c) material in which atom move away from each other and with size in the range of 1 to 100 nano meters.

d) material in which atom move away from each other and with size in the range of 1 to 50 nano meters.

Ans1: (b) material in which atom move away from each other and with size in the range of 1 to 50 nano meters.

Hint: Can be found inside Q1 on page number 1 in the notes.

Q.2. Which of the following materials can be used to formed nanomaterials?

a) sand

b) paper

c) polymers

d) none of these

Ans2: (c) polymers.

Hint: Can be found inside Q1 on page number 1 in the notes.

Q.3. What are two approaches in nanotechnology?

a) Bottom up approach & Top down approach

b) Bottom down approach & Top down approach

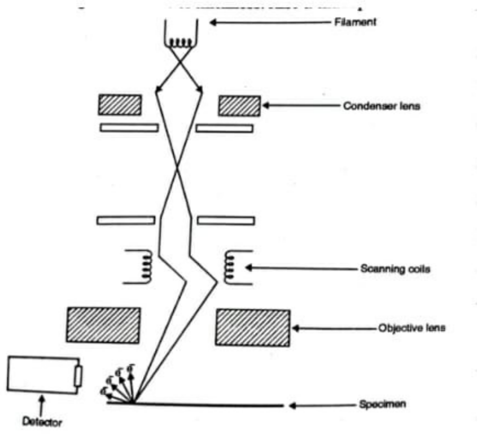
c) Bottom down approach & Top up approach

d) Bottom up approach & Top up approach

Ans3: (a) Bottom up approach & Top down approach.

Hint: Can be found inside Q2 on page number 1 in the notes.

Q.4. Identify what type of diagram this is?



a) Scanning Electron Microscope

b) Atomic Force Microscope

c) Ball Milling method

d) Sputtering method

Ans4:(a) Scanning Electron Microscope

Hint: Can be found inside Q.3 on page number 2 in the notes.

Q.5. How nanomaterials are formed in Bottom up approach?

a) By building atom by molecule or molecule by molecule.

b) By building atom by atom or molecule by atom.

c) By building atom by atom or molecule by molecule.

d) none of these

Ans5: (c) By building atom by atom or molecule by molecule.

Hint: Can be found inside Q2 on page number 1 in the notes.

Q.6. What is Scanning Electron Microscope(SEM)?

a) SEM is used to obtain the power of an electron.

b) SEM is used to obtain mass of the microscope.

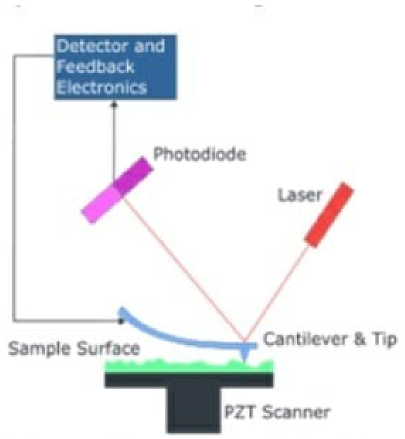
c) SEM is used to obtain the charge of an electron.

d) SEM is used to obtain images of surface of thickness.

Ans6: (d) SEM is used to obtain images of surface of thickness.

Hint: Can be found inside Q3 on page number 2 in the notes.

Q.7. Identify what type of diagram this ?



a) Scanning Electron Microscope
b) Atomic Force Microscope
c) Ball Milling method
d) Sputtering method

Ans7:(b) Atomic Force Microscope

Hint: Can be found inside Q.4 on page number 3 in the notes.

Q.8. What is the name of the gun from which electron beam is obtained?

a) laser gun

b) electron gun

c) hot glue gun

d) none of these

Ans8: (b) electron gun.

Hint: Can be found inside Q3 on page number 2 in the notes.

Q.9. How much size of specimen can be clearly resolved by SEM?

a) 1000Ao

b) 2000Ao

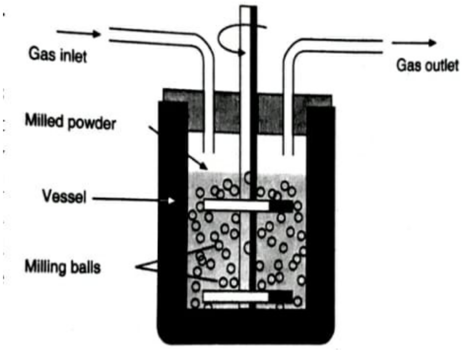
c) 50Ao

d) 500Ao

Ans9: (c) 50Ao.

Hint: Can be found inside Q3 on page number 2 in the notes.

Q.10. Identify what type of diagram this is ?



a) Scanning Electron Microscope
b) Atomic Force Microscope
c) Ball Milling method
d) Sputtering method

Ans10:(c) Ball Milling method

Hint: Can be found inside Q.5 on page number 4 in the notes.

Q.11. Which is the most versatile and powerful microscopy technology for studying samples in nano-scale?

a) Atomic Force Microscopy

b) SEM

c) nanotechnology

d) none of these

Ans11: (a) Atomic Force Microscopy.

Hint: Can be found inside Q4 on page number 3 in the notes.

Q.12. In which process a powder mixture placed in the ball mill is subjected to high energy collision from the balls?

a) Ball Milling Method of nano particle synthesis

b) AFM

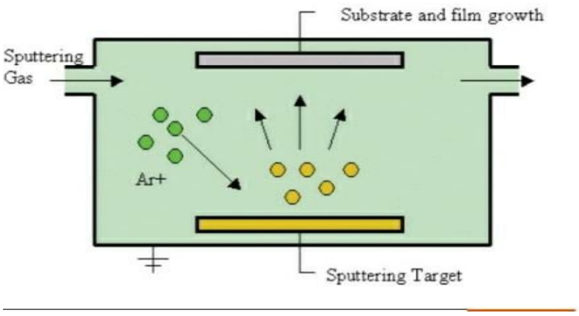
c) SEM

d) surface sensing

Ans12: (a) Ball Milling Method of nano particle synthesis.

Hint: Can be found inside Q5 on page number 4 in the notes.

Q.13. Identify what type of diagram this is?



a) Scanning Electron Microscope
b) Atomic Force Microscope
c) Ball Milling method
d) Sputtering method

Ans13:(d) Sputtering method

Hint: Can be found inside Q.6 on page number 4 in the notes.

Q.14. which material of the inner suface of shell is used in Ball Milling Method?

a) water resistant

b) abrasion resistant

c) dust resistant

d) all of the above

Ans14: (b) abrasion resistant.

Hint: Can be found inside Q5 on page number 4 in the notes.

Q.15. Due to what argon atoms get ionized?

a) constant acceleration

b) charge density

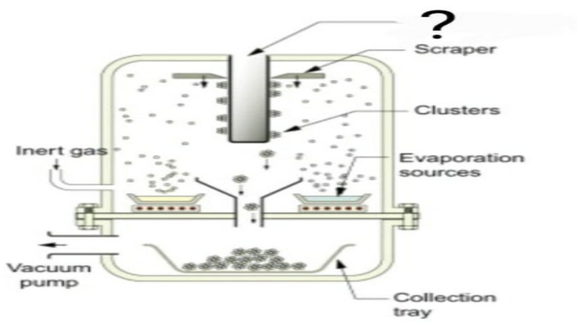
c) gas pressure

d) large potential difference

Ans15: (d) large potential difference.

Hint: Can be found inside Q6 on page number 5 in the notes.

Q.16. Which part is missing in the given diagram?



a) Sputtering gas
b) Cold finger
c) Gas inlet
d) Gas outlet

Ans16:(b) Cold finger

Hint: Can be found inside Q.7 on page number 5 in the notes.

Q.17. How thickness of the film can be controlled?

a) by varying acceleration

b) by varying potential difference

c) by varying mass

d) by varying argon gas pressure

Ans17: (d) by varying argon gas pressure.

Hint: Can be found inside Q6 on page number 5 in the notes.

Q.18. What is cold finger?

a) the liquid nitrogen cooled cylinder

b) cooled tank

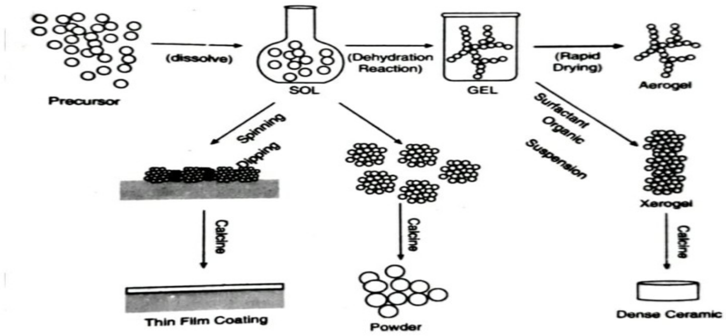
c) scraper

d) none of these

Ans18: (b) cooled tank

Hint: Can be found inside Q5 on page number 4 in the notes.

Q.19. Identify which type of diagram this is?



a) Scanning Electron Microscope
b) Atomic Force Microscope
c) Ball Milling method
d) SOL-Gel Technique method

Ans19:(d) SOL-Gel Technique method

Q.20. How nanomaterials can be generated in hydra dynamic cavitation?

a) through exploding waste material

b) through creation and release of gas bubbles

c) through opening the mouth of the cooled tank

d) none of these

Ans20: (b) through creation and release of gas bubbles.

Hint: Can be found inside Q8 on page number 6 in the notes.

SUGGESTION

1.Different methods of PDV need to be added.

2.Construction and working of AFM need to be added